Does Punishment Matter? A Refinement of the Inspection Game

RIMAWAN PRADIPTYO
University of York

We attempt to revise the inspection game used by George Tsebelis to model phenomena in criminal justice. The refinement has been conducted by disaggregating the game payoffs and then using findings from empirical studies to reconstruct the game. In contrast to Tsebelis' propositions, we find that the severity of punishment may affect the offending behavior of individuals. The result also holds for the case in which the authority initiates crime prevention programs, by providing incentives to those who do not have a criminal history. The impact of increasing the severity of punishment on reducing individuals' offending behavior is less certain than that of instigating crime prevention programs. This result holds so long as the authority does not alter the levels of enforcement.

1. INTRODUCTION

The effectiveness of sentencing as an instrument to deter crime has long been debated. Not only have such debates been conducted on a theoretical level, but empirical[1] and experimental studies[2] have yet to produce a unified view on the effectiveness of sentencing. So far, economists have analyzed individuals' offending behavior based on decision theory and game theory. The former approach was used in a seminal paper by Gary Becker (1968)[3] and the latter by Tsebelis (1989).[4]

Tsebelis argued that the interaction of agents in criminal justice is better analyzed using game theory. Based on this approach, the probabilities of success and failure in committing an offense are affected by the interactions of rational players. He analyzed phenomena in criminal justice as a one-shot 2×2 inspection game, played simultaneously.[5] The results from Tsebelis' (1989, 1990, 1991, 1992, 1993; Bianco/Ordeshook/Tsebelis, 1990[6]) studies contradict Becker's propositions. The former asserts that punishment is not effective in influencing the tendency of individuals or firms (countries) to

commit illegal activities (to breach international law). Any attempts to increase the severity of punishment are counter-productive, since individual's or firm's behavior will not be affected; instead, the policy adversely affects the intensity with which the authority can impose the law.[7] Although Tsebelis (1991, 1992) used the same game to analyze phenomena in the areas of industrial policy and international relations, it is beyond the scope of this paper to discuss the application of the analysis in both fields.

This paper aims to refine the inspection game proposed by Tsebelis (1989, 1990, 1993; BOT, 1990). In section 2 we discuss the comparability between Becker's and Tsebelis' models. Apart from the use of different analytical tools, we identify several factors which should be taken into consideration in comparing these models.

A refinement of the inspection game is presented in Section 3. This has been done by disaggregating the game payoffs and then using findings from empirical studies to reconstruct the game. Our working assumption is that the severity of punishment has a positive correlation with the costs of delivering court sentences. Several studies provide evidence for this assumption.[8]

We found that an increase in the severity of punishment will reduce the likelihood of enforcement of the law. This result holds so long as the policy increases the individuals' disutility of serving court sentences, and it does not increase the immediate utility of offending. In addition, there is a condition in which an increase in the severity of punishment reduces the likelihood of offending (see Section 4).

In section 5 we discuss the impact of crime prevention initiatives on individuals' offending behavior. Instead of increasing the severity of punishment, the authority may theoretically provide financial incentives for those who do not have criminal records. In the real world, these policies have been implemented by providing support to a group of individuals who are at risk of offending. The results show that the implementation of crime prevention initiatives, cet*eris paribus*, is going to reduce the probabilities of both offending and enforcing the law.

2. COMPARABILITY BETWEEN THE TSEBELIS AND BECKER MODELS

The main contrast between the Tsebelis (1989, 1990, 1993; BOT, 1990) and Becker (1968) models[9] lies in the effectiveness of criminal justice interventions in deterring or preventing individuals from criminal careers.

The use of different analytical tools is attributable to the discrepancies in results between both models, as Tsebelis (1989:84) states:

> *The logical and empirical evidence in favor of the game-theoretic approach indicates that both dominant approaches to crime prevention—the sociological and the economic—are partial and short-run in nature. They are both based on the assumption that modifying the incentive structure of a rational agent* (*modifying payoffs*) *will affect that agent's behavior. Reasonable and plausible as this assumption may seem, it was shown not to withstand game-theoretic scrutiny.*

Individuals' offending behavior has been modelled as a one-shot 2×2 game between public and police, played by representative agents. The payoff matrix of the game is as follows:

TABLE 1: **Game 1**

		Police	
		Enforce	Not Enforce
Public	Offend	a_1, a_2	b_1, b_2
	Not Offend	c_1, c_2	d_1, d_2

where: $c_1 > a_1$, $b_1 > d_1$, $a_2 > b_2$, and $d_2 > c_2$.

The game does not have a pure strategy equilibrium; instead it has a unique mixed strategy equilibrium, which implies that a total reduction in the number of offenses is impossible. Let p be the probability of the public to offend and q be the probability of the police to enforce the law. The mixed strategy equilibrium of the game is as follows (theorem 1 of Tsebelis, 1989):

$$p^* = \frac{d_2 - c_2}{a_2 - b_2 - c_2 + d_2} \qquad 1$$

$$q^* = \frac{b_1 - d_1}{b_1 - d_1 - a_1 + c_1} \qquad 2$$

Tsebelis argues that any attempt to increase the severity of punishment will alter the payoffs only for individuals, namely $a_1' < a_1$ and $c_1 > a_1'$. This policy leaves unchanged the frequency of violation at equilibrium (p^*) (theorem 2 of Tsebelis); however, it decreases the likelihood of enforcing the law (q'^*) (theorem 3 of Tsebelis). Hirshleifer and Rasmusen (1992) labelled these results as the payoff irrelevance proposition (PIP).[10] A similar result will be obtained if the authority attempts to increase either c_1 or d_1 (welfare

measures) (theorem 4 and 5 of Tsebelis). Tsebelis claims that the results are robust in relation to any alterations in the settings.[11]

Tsebelis' argument that the offending behavior of individuals is more appropriately analyzed using game theory rather than decision theory has been widely accepted and appreciated. Nevertheless, his controversial proposition on the ineffectiveness of punishment has attracted many critics (see BOT, 1990; Hirshleifer and Rasmusen, 1992; Weissing and Ostrom, 1991). Most of the comments focus on proving that the PIP phenomena applies only in certain conditions–for instance, if the game is played by no more than two players (BOT, 1990), if the game with discrete payoffs is played simultaneously, or if the game is played sequentially with the public move being first (Weissing and Ostrom, 1991).

Despite the overwhelming criticism, few attempts have been made to comment on the construction of the game, even though the most crucial factor supporting Tsebelis' propositions lies in the specification of the game. The PIP is valid if the analysis has been conducted for a generic game; however, the model may not necessarily accurately represent phenomena in criminal justice. The conclusions drawn from applying a game theoretical model in criminal justice depend on the degree to which the model itself represents real phenomena in that field.

The discrepancies between the Tsebelis and Becker models do not arise solely due to the use of different analytical tools. Indeed, the discrepancies are mostly attributable to the fact that the approaches start with different assumptions. It should be noted that the following list of discrepancies applies also to more recent studies based on Becker's approach, such as Polinsky and Shavell (2000, 2007). Several factors which make the models non-comparable are as follows:

1. *Players and their roles*. There are two players in Becker's model, namely individuals and a criminal justice authority (CJA), in which police are a part of the organization. The CJA determines the levels of punishment and enforcement of the law. It is clear, therefore, that the CJA finances the delivery of sentences and also the enforcement of the law. In Tsebelis' models, however, the police seem to be an independent institution and there is no clear indication whether the police are part of the CJA.

2. *Endogeneity of punishment*. In Becker's model (and also in Polinsky and Shavell, 2000, 2007), the levels of punishment and enforcement of the law are determined endogenously by CJA. In contrast, the level of

punishment is an exogenous factor in Tsebelis' models. Attempts have been made to endogenize this variable by introducing a legislature as the third player (Tsebelis, 1990; BOT, 1990), however, the legislature does not have the authority to determine the level of enforcement.

3. *Sequence of play*. In Becker's approach, as in Polinsky and Shavell, it is implicitly assumed that the authority moves first, by setting the level of enforcement and severity of punishment, and is then followed by individuals (whether or not they offend). In Tsebelis' models however, players play the game simultaneously.

4. *Objective function*. In Becker's model, an individual maximizes his own payoff, but the authority maximizes the social welfare function.[12] This model captures the interaction between a rational and selfish individual with a benevolent government. As a consequence, the CJA influences individuals' behavior by finding the optimum level of punishment and enforcement which minimizes social costs. In the Tsebelis models, on the other hand, both players maximize their own utility.[13] This approach is in line with the literature on public choice. In this model, the police are not concerned with the net benefits of altering punishments. It is obvious, therefore, that any attempts to increase the severity of punishment will not affect the police payoff.

Any attempt to compare the models should alter one of the constructions. This can be approached from several avenues. Firstly, one might analyze the case as a two-stage game. In the first stage, the CJA sets the level of punishment and also the budget for enforcing the law. In the second stage, given the level of punishment and the budget available, the police and the public play the game.

Secondly, the game can be maintained as a 2-player 2×2 game; however, the police should be replaced by the CJA. In this setting, the CJA sets the level of punishment and enforcement of the law. It is assumed that the police are a part of the CJA, and the CJA determines performance indicators for the police. Both the severity of punishment and the intensity of enforcing the law are considered as endogeneous variables. In this scenario, there is no justification for separating the role of the police and the CJA, since the police are a component of the CJA.

Finally, a combination of both of the above approaches can be used. In addition, we assume that any criminal justice interventions are costly. The

implication of these settings is that any attempt to increase the level of punishment or to initiate crime prevention initiatives will affect both players' payoffs.

3. THE REVISED INSPECTION GAME

We model phenomena in criminal justice as a 2-player 2×2 one-shot game played by representative agents, namely public and enforcer. It is assumed that the enforcer is a broader institution than the police, yet the enforcer is part of a higher organization, namely the CJA. The CJA finances the enforcer and it has the authority to set the level of punishment. Given the punishment regime, the enforcer has the duty of enforcing the law and delivering criminal justice interventions, including sentencing. This setting is consistent with Tsebelis' models, which assumed that the level of punishment is exogenous, and at the same time it accommodates Becker's model by incorporating the allocation of resources by the enforcer in tackling crime.

The interactions between the CJA and the enforcer can be analyzed using the principal-agent theory. The CJA, as a principal, has the authority to set objectives or goals which should be fulfilled by the enforcer. These goals can be used as performance measures against the enforcer's efforts. It is apparent that the level of enforcement by the enforcer depends on guidance and financial support from the CJA. In the U.K., for instance, the Secretary of State for Home Affairs publishes a national policing plan every three years which sets targets for national policing. Accordingly, each police force at regional level should develop their own plan, which conforms to the national plan and, at the same time, accommodates local issues (Home Office, 2002).

Ideally, the interaction between public and the enforcer in the area of criminal justice should be analyzed as a repeated game. The disutility of being convicted, for instance, is not limited only to serving *direct punishment* (e.g., paying a fine or staying in prison) but, more importantly, there is a substantial reduction of potential future wealth owing to loss of reputation (we define this as an *indirect punishment*). This negative reputation effect is similar to the impact of not obtaining qualifications on an individual's future income. If we pursue the analysis in a one-shot game, these reputation effects should be taken into consideration in the model.

A further modification from the Tsebelis model lies in the specification of the payoffs. In the original model, each element of the payoffs (i.e., a, b, c and d) represents the net benefits of choosing a strategy, given the strategy taken by the opponent. In this model, we provide the identity of each

element in the payoff matrix and the game is given as follows:

TABLE 2: Game 2

		Enforcer: Enforce	Enforcer: Not Enforce
Public	Offend	$U_O - U_D$, $B_E - C_E - C_S$	$U_O + U_R$, 0
	Not Offend	U_R , $B_R - C_E$	U_R , B_R

where:

U_O = immediate utility arising from committing an offense.

U_D = disutility of serving direct punishment (e.g., imprisonment, fine, community service).

U_R = positive reputational effects for individuals of not being convicted.

B_E = benefits of enforcing the law, including the detection of incidents and any deterrence effects which arise due to enforcement of the law.

B_R = reputational benefits in achieving objectives set by the CJA.

C_E = costs of enforcement of the law, including, for instance, costs of investigation and of dispatching police officers to certain areas.

C_S = costs of delivering court sentences, including direct and indirect punishments.

Individuals who have never been arrested or convicted receive benefits in terms of positive reputation effects (*reputation effects* hereafter) U_R. Parts of this effect are: easy access to financial services (e.g., mortgages, interest-free credit); freedom to travel; easy access to the labor market; and the respect and trust of friends, neighbors and colleagues. This effect does not apply, however, to individuals who are convicted, whose access to the job market,[14] future wealth, and even mobility[15] may be adversely affected by stigma or the loss of reputation effects ($-U_R$).[16]

From the public's perspective, committing an offense produces immediate benefits U_O, either in terms of material well-being or psychological rewards (e.g., admiration from other criminals or relief from peer pressures). By the same token, this activity produces disutility to individuals (U_D) if they are convicted and have to serve court sentences. The longer (higher) the imprisonment term (fines), the higher the disutility of serving direct punishment (U_D). The disutility of serving direct punishment ranges from the loss of earnings to the loss of liberty due to serving a sentence.

Our working assumption is that the disutility of serving a direct punishment

(U_D) is independent of stigma or the loss of reputational effects ($-U_R$) following conviction. As a consequence, there are three possible strategies for the CJA to increase the severity of punishment: a) increasing the severity of direct punishment (U_D), for instance by increasing the period of imprisonment or by increasing the value of fines levied; or b) increasing the severity of indirect punishment, which can be achieved, for instance, by extending the period for keeping individual criminal records, by extending types of positions which cannot be taken by ex-offenders, or by extending the period of probation during which offenders must report their mobility to police after their release; or c) a combination of both approaches.

If an individual decides to offend, irrespective of whether the law has been enforced or not, he/she will earn immediate utility of offending (U_O). In the case in which the law has not been enforced, an offender enjoys the immediate utility of offending (U_O), while keeping intact his or her positive reputational effects (U_R). In the case in which an individual commits an offense and the law has been enforced, the offender will earn immediate utility of offending (U_O), but at the same time, he/she will have to bear the disutility of serving direct punishment (U_D). If the individual decides not to offend, irrespective of whether the law has been enforced or not, he/she will be able to keep positive reputational effects (U_R).

It is the duty of the CJA to finance the enforcer in delivering court sentences and enforcing the law. Enforcing the law is costly (C_E), and so is delivering sentences (C_S). Each type of sentence inflicts transaction costs, especially for policing or maintaining the order of disposals. Suppose the law has been enforced, regardless of the actions of individuals, the enforcer incurs costs to enforce the law (C_E). A good example of enforcement costs is the monies spent by both the U.S. and U.K. governments on conducting operations aimed at tracing pedophiles. The costs of conducting such operations are separate from the costs of keeping pedophiles in prison or of monitoring ex-offenders after trelease (C_S).

Suppose individuals commit an offense and the law has been enforced, there will be benefits of enforcing the law (B_E). This variable includes the ability of the enforcer to detect the crime and, subsequently, to send offenders to court, the possible recovery of some of the victims' materials from offenders, and benefits arising from the sentences following conviction. The last element has been derived from the fact that different sentences produce different benefits.

In general, the objectives of sentences are: a) general deterrence—providing a signal to potential offenders not to commit an offense; b) specific

deterrence—deterring re-offense in the future; c) punishment; d) rehabilitation; e) incapacitation—isolating offenders from the rest of society during the serving period; and f) restitution—restoring the losses incurred by victims (Bowles and Pradiptyo, 2004). Each outcome may have a different content of benefits. Fines, for instance, do not have an element of incapacitation, whereas imprisonment obviously does.

It is the enforcer's duty to achieve objectives determined by the CJA. In the case where these objectives are met, the enforcer earns positive reputational benefits (B_R) which definitely will be obtained if, irrespective of the strategies chosen by the enforcer, the public decides not to offend. Suppose, however, that individuals commit an offense, but the enforcer decides not to enforce the law, then the crimes will be undetected. In this case, the enforcer will not bear any costs, however, they will lose benefits B_E.

Consider q is the probability that the enforcer will enforce the law. From the individual's perspective, he/she will commit an offense if :

$$(U_O - U_D)\,q + (U_O + U_R)(1-q) \geq U_R q + U_R(1-q)$$

$$U_O \geq q\,(U_D + U_R) \qquad 3$$

A similar method is used by the enforcer to decide whether to enforce the law. Consider p to be the probability of the public to offend, thus the law will be enforced if:

$$(B_E - C_E - C_S)\,p + (B_R - C_E)(1-p) \geq B_R(1-p)$$

$$pB_E \geq C_E + p\,C_S \qquad 4$$

Similar to Tsebelis' results, the game above does not have a pure strategy equilibrium. The mixed strategy equilibrium is as follows:

$$p^* = \frac{C_E}{B_E - C_S} \qquad 5$$

$$q^* = \frac{U_O}{U_D + U_R} \qquad 6$$

Since $p^*, q^* \in (0,1)$ it can be inferred that the underlying assumptions of the model are as follows:

$$B_E > C_E + C_S$$

$$U_D + U_R > U_O$$

Proposition 1: In a 2×2 inspection game as specified in Table 2:

1. An individual will commit an offense if the utility to conduct such activity dominates the expected disutility of serving direct punishment and the expected loss of reputational effects (i.e., equation 3);

2. The law will be enforced if the expected benefits of enforcing it dominate the costs of enforcement and the expected costs of delivering sentences (i.e., equation 4);

3. In equilibrium, the probability to offend is a reverse function of the benefit:cost ratio of enforcing the law minus the ratio of enforcement and sentencing costs (i.e., equation 5);

4. In equilibrium, the probability of enforcing the law has a positive correlation with the individual's utility to offend, but it has negative correlation with the disutility of serving direct and indirect punishments due to conviction (i.e., equation 6).

Another interpretation of equation (5) is that, given the level of enforcement (i.e., C_E), the probability of an individual committing an offense increases (decreases) as the net benefits of setting the severity of punishment (i.e., $B_E - C_S$) decrease (increase). In order to influence individuals' offending behavior, it is crucial for the enforcer to implement types of punishments (either direct or indirect) which have positive net benefits. Equation (6), on the other hand, implies that if the enforcer observes that an increase in severity of punishment increases either U_D or U_R or both, there is no incentive for the enforcer to increase or to maintain the level of enforcement. The following sections discuss the impact of an increase in the severity of punishment and the instigation of crime prevention programs on reducing offending behavior and enforcement of the law.

4. INCREASING THE SEVERITY OF PUNISHMENT

Throughout this article, we assume that there is a positive correlation between the severity of punishment and the costs of delivering sentences (*CS*). The correlation is stronger in countries where capital punishment has been abolished. The longer the period for which an individual is sentenced to either imprisonment or community service, the more expensive the costs

incurred by the government. In England and Wales, for instance, the average cost to keep an offender in prison is approximately £30,000 per annum, whereas the cost of community sentence per offender is about £3,000 per annum (Bowles and Pradiptyo, 2004; Bowles et al., 2004).

Many economists in the area of crime economics share the belief that fines are the most favorable punishment, as Becker (1968:180) argued:

> *The total social cost of punishments is the cost to offenders plus the cost or minus the gain to others. Fines produce a gain to the latter that equals the cost to offenders, minus collection costs, and so the social cost of fines is about zero, as benefits a transfer payment. The social cost of probation, imprisonment, and other punishments, however, generally exceeds that of offenders, because others are also hurt.*

European countries tend to use fines more intensively than other developed countries such as the U.S. and Israel. In 1986, for instance, 81% of adult offenders in Germany were ordered to pay fines (Tonry, 1997). In addition, fine payment covers 91% of court dispositions in Finland in 1979. In the U.S., however, fines have been used more often as an additional penalty, combined with a primary penalty of either imprisonment or community service, rather than as a sole sanction. In year 2000, of 28,810 convicts in the 75 largest counties, only 1% received a fine as a sole sanction (U.S. DOJ, 2003). Similarly, Einat (2004) reported that in Israel during 1997-2000 fines were used as a sole sanction for only 11% of cases on average.

Empirical studies show that the costs to collect fines are substantial and may increase as the value of fines increases. According to Chapman et al. (2002), the costs of collecting fines in the U.K. is almost one-third of the value of the fines. This applies when the average value of the fine is £200, which is much lower than the average social costs of crime.[17] In addition, variations in the costs to collect fines in the U.K. are high, ranging from 11 pence to 44 pence per pound collected (DCA, 2004).

The effectiveness of fines as a better instrument for imposing sanction, as opposed to other dispositions, is under scrutiny. In particular this is related to the low payment rate. Fines have a deterrence effect only when they are paid by offenders. According to Einat (2004), 28 out of 40 (70%) of magistrate judges in Israel[18] who participated in a study thought that the source of fine ineffectiveness is the fact that most offenders are poor and thus unable or reluctant to pay fines. In the U.S., the evaluation of a pilot project on day-fines in Staten Island in 1988-1989 showed that only 70% of offenders paid their fines in full (Tonry, 1997, quoting a 1992 article by Hillsman and Greene). In the U.K., the rate of fine payment is about 55% in England

and Wales (DCA, 2004). If this disposition rate were translated into imprisonment, the result would be as if 45% of prisoners escaped (Bowles and Pradiptyo, 2004).

The ineffectiveness of a small fine was reported by Gneezy and Rustichini (2004), based on findings from their experiment. The results show that imposing a small fine fails to reduce the unwanted behavior. On the contrary, it increases the tendency to repeat the behavior for which fines were imposed. They argue that a fine will reduce the unwanted behavior only if it is imposed on a large scale. While this argument seems appealing, in the area of criminal justice there is an upper limit on the fines imposed, which is the wealth of offenders. A large fine cannot be effectively imposed upon offenders with low incomes or those who receive unemployment benefits. The effectiveness of fines as a deterrence instrument may be limited, since the majority of offenders come from low income backgrounds.

In order to make fines more credible, many authorities have to adopt a strategy of transforming the values of fines in relation to a term of imprisonment period—as a result, a failure to pay the fines will be compensated by serving time in prison. In the U.S., for instance, 25% of convicts sentenced by state courts in the year 2000 received fines as additional penalties (U.S. DOJ, 2003). In Israel during 1997-2000, fines were used in combination with other penalties in 34.7% of the cases (Einat, 2004). The use of complementary sanctions shows that fines in themselves are not sufficient as a credible sentence. Furthermore, the costs of policing and enforcing fines may not necessarily be lower than other types of sentences, and the higher the fine, the higher the costs of enforcing and policing it. Any attempt to increase the value of fines may increase the number of defaults. In turn, this gives rise to an increase in the number of inmates in prison. As a result, fines may not be a good solution when tackling overcrowded prisons.

As discussed in the previous section, there are three possible ways to increase the severity of punishment:

1. Increasing the disutility of serving court sentences (U_D), which can be done by increasing the duration of imprisonment or by increasing the fines levied upon offenders.
2. Increasing the loss of offenders' reputational benefits (U_R); or
3. A combination of both methods.

Suppose the CJA decides to increase the severity of direct punishment, *ceteris*

paribus. This policy affects the elements of payoffs $U_D, C_S,$ and B_E such that $U'_D > U_D, C'_S > C_S$, and $B'_E > B_E$. This implies that the policy affects payoffs (offend, enforce) as follows:

$$(U_O - U'_D) < (U_O - U_D) \quad 7$$

$$B'_E - (C_E + C'_S) \gtreqless B_E - (C_E + C_S) \quad 8$$

Since $U'_D > U_D$, equation (7) shows that an increase in the severity of direct punishment reduces offenders' net benefits of offending. Nevertheless, the enforcer may not necessarily be in favor of the policy if its benefits are less than its costs. Equation (8) shows that the impact of the policy on the enforcement payoff is uncertain when the public commits crime. Although we obtain that $C'_S > C_S$, and $B'_E > B_E$, the relationship between C'_S and B'_E cannot be determined.

So long as the CJA fully compensates any additional costs incurred by the enforcer, an increase in the severity of punishment will not change the benefits of achieving the CJA's objectives (B_R). The policy does not change the costs of enforcing the law (C_E), however, it does increase the costs of delivering sentences ($C'_S > C_S$), and in turn it increases the benefits of enforcement ($B'_E > B_E$).

The second route that can be pursued by the CJA is to increase the loss of reputational effects (U_R) borne by offenders who are convicted, given the level of disutility in serving court sentences (U_D). This can be done, for instance, by extending the probationary period to ex-offenders, extending the list of positions that cannot be taken by ex-offenders, or extending the period in which ex-offenders have to report to police when they make a journey. In this case, the decision of individuals to offend means that they are risking a higher stake with regard to their reputation. From the public's perspective, this policy affects their payoffs as follows:

$$U_O + U'_R > U_O + U_R \quad 9$$

$$U'_R > U_R \quad 10$$

From the enforcer's perspective, this policy entails an increase in policing costs $C''_S,$ and in the benefits of imposing the policy $B''_E,$ such that $C''_S > C_S,$ and $B''_E > B_E,$ thus we obtain:

$$B''_E - (C_E + C''_S) \gtreqless B_E - (C_E + C_S) \quad 11$$

The final method of increasing the severity of punishment can be pursued by

combining both policies. This policy affects U_D, U_R, C_S and B_E such that $U'_D > U_D, U'_R > U_R, \hat{C}_S > C_S$ and $\hat{B}_E > B_E$, where $\hat{C}_S = C'_S + C''_S$. Indeed, the benefit:cost ratios of this policy may be different from those of the previous policies. Nevertheless, similar to the previous policies, we obtain the following relationship:

$$\hat{B}_E - (C_E + \hat{C}_S) \gtreqless B_E - (C_E + C_S) \qquad 12$$

Although we can determine that $\hat{B}_E > B_E$ and $\hat{C}_S > C_S$, it cannot be determined that the net benefit of the new policy is either higher, lower, or the same as that of the old policy. Game 3 depicts the payoff matrix of the new game, whereby the respective payoffs in Game 2 are substituted by equations (7), (9), (10), and (12).

TABLE 3: Game 3

		Enforcer	
		Enforce	Not Enforce
Public	Offend	$U_O - U'_D, \hat{B}_E - (C_E + \hat{C}_S)$	$U_O + U'_R, 0$
	Not Offend	$U'_R, B_R - C_E$	U'_R, B_R

Let $\hat{p}^*$ and $\hat{q}^*$ be the new equilibrium under the new policy, given as follows:

$$\hat{p}^* = \frac{C_E}{\hat{B}_E - \hat{C}_S} \qquad 13$$

$$\hat{q}^* = \frac{U_O}{U'_D + U'_R} \qquad 14$$

where $\hat{p}^*, \hat{q}^* \in (0,1)$.

The policy may not necessarily reduce the likelihood of individuals to offend. So long as the marginal benefit:cost ratio of the new policy is undetermined, there is no guarantee that the policy reduces the likelihood of individuals to offend. This policy, however, reduces the likelihood of enforcement of the law ($\hat{q}^* < q^*$). This occurs since $U'_R > U_R$ and there is no impact by the policy on individuals' immediate utility to offend (U_O). On the contrary, there are three possibilities for the impact of the policy upon the probability of offending—namely, decrease, increase or remain constant. The following conditions apply:

$\hat{p}^* < p^*$ if:

$$(\hat{B}_E - \hat{C}_S) > (B_E - C_S) \quad 15$$

and $\hat{p}^* \geq p^*$ if :

$$(\hat{B}_E - \hat{C}_S) \leq (B_E - C_S) \quad 16$$

The likelihood of individuals offending decreases (increases) if the individuals observe that, given the level of enforcement, from the enforcer's perspective the net benefit of an increase in the severity of punishment (U_D) is larger (smaller) than that of the initial level. Similar results are obtained for the cases in which the CJA decides to increase either U_D or U_R separately.

> **Proposition 2**: An increase in the severity of punishment reduces the probabilities of enforcing the law and of offending, so long as the net benefits of enforcing the new policy dominate those of the initial policy.

Indeed, there are various methods available to increase the severity of punishment; however, it does not necessarily follow that the benefits of each method dominate its costs. It is imperative, therefore, for the enforcer to choose interventions in which, from the enforcer's perspective, their benefits dominate their costs. This can be done by adopting interventions similar to ones implemented elsewhere which have already proven their benefit:cost ratios are larger than 1. If this can be observed by individuals, or the enforcer sends a signal to the public by announcing the benefit:cost ratio of the interventions, then the enforcer may be able to influence the offending behavior of individuals.

Up to this point, the results suggest that any attempt to increase the severity of punishment reduces the likelihood of enforcement of the law. The results hold so long as we assume that any increase in the severity of punishment affects either U_D, U_R or both, and it does not affect individuals' immediate utility of offending (U_O).

We extend the analysis by assuming that the utility of offending (U_O) may also increase as the severity of punishment increases. The more severe the punishment given to drug traffickers, for instance, the more expensive becomes the selling price of drugs, which may lead to an increase in profit from the activities. So long as there are individuals who perceive that the expected benefits of an illegal activity may exceed their expected costs, there will be temptations to conduct such activity. There is also a tendency for offenders to gain more admiration from their peers if they commit a crime

with more severe punishment. Suppose we apply this assumption into Game 3 above; the corresponding payoff matrix of the new game is as follows:

TABLE 4: Game 4

		Enforcer	
		Enforce	Not Enforce
Public	Offend	$U'_O + U'_D, \hat{B}_E - (C_E + \hat{C}_S)$	$U'_O + U'_R, 0$
	Not Offend	$U'_R, B_R - C_E$	U'_R, B_R

where: $(U'_O + U'_D) \gtreqless (U_O + U_D)$ and $U'_O + U'_R > U_O + U_R$.

The new mixed strategy equilibrium of Game 4 is as follows:

$$\overset{\circ}{p}^* = \frac{C_E}{\hat{B}_E - \hat{C}_S} \quad 17$$

$$\overset{\circ}{q}^* = \frac{U'_O}{U'_D + U'_R} \quad 18$$

where $\overset{\circ}{p}^*, \overset{\circ}{q}^* \in (0,1)$.

Slightly different from the results in Game 3, under the new assumption it cannot be determined that the likelihood of enforcing the law decreases due to the implementation of the policy. There are three possible impacts of the policy on the likelihood that the law will be enforced. $\overset{\circ}{q}^* < q^*$ if:

$$\frac{U'_O}{U'_D + U'_R} < \frac{U_O}{U_D + U_R}$$

Since $U'_O = U_O + \Delta U_O, U'_D = U_D + \Delta U_D, U'_R = U_R + \Delta U_R$, and ΔU_O, ΔU_D, $\Delta U_O \in R^+$, substituting the respective values and re-arranging the equation above, $\overset{\circ}{q}^* < q^*$ if:

$$\Delta U_O < \Delta U_D + \Delta U_R \quad 19$$

Using a similar method, $\overset{\circ}{q}^* \geq q^*$ if the following condition holds:

$$\Delta U_O \geq \Delta U_D + \Delta U_R \quad 20$$

So long as an increase in U_R or both U_D and U_R is followed by an increase in U_O, and an increase in U_O dominates either an increase in U_R or both U_R and U_D, there will be an increase in the probability of enforcing the law.

Proposition 3: An increase in the severity of punishment may increase the probability of enforcing the law if the policy increases individuals' utility of offending and the change in utility of offending dominates the change in disutilities of conviction.

5. CRIME PREVENTION INITIATIVES

Crime prevention initiatives are alternative methods of reducing the number of offenses by preventing individuals from pursuing a criminal career. In the most extreme case, this program can be interpreted as distributing financial incentives to individuals who do not have criminal records. Theoretically this intervention is feasible, even though in reality, crime prevention initiatives have mostly been conducted by providing support to a group of individuals who are at risk of committing an offense. Early intervention initiatives in criminal justice that target children and youths at risk can be classified as part of these initiatives. Examples of such programs are the Perry preschool (Karoly et al., 2001) and Abecedarian preschool (Clarke and Campbell, 1998) in the U.S., and also the Youth Inclusion Programme (YIP) (Youth Justice Board, 2003) and the Safer School Partnership (SSP) (Youth Justice Board, 2005) in the U.K.

If we suppose the CJA decides to instigate crime prevention programs by providing financial rewards to those who do not have criminal histories, the following may obtain. From the individuals' perspective, the policy entails that those who are not convicted are entitled to receive a transfer payment (denoted by T_R, such that $T_R > 0$). In other words, only those who offend and are convicted do not receive the transfer payment. From the enforcer's perspective, the total value of transfers given to individuals is equal to the costs of the initiatives (denoted by C_P, such that $C_P > 0$). The payoff matrix of the new game is as follows:

TABLE 5: **Game 5**

		Enforcer	
		Enforce	Not Enforce
Public	Offend	$U_O - U_D,\ B_E - C_E - C_S$	$U_O + T_R + U_R,\ -C_P$
	Not Offend	$T_R + U_R,\ B_R - C_E - C_P$	$T_R + U_R,\ B_R - C_P$

If the enforcer decides not to enforce the law and the public commit

offenses, not only do the offenders enjoy the utility of committing offenses (U_O) while keeping intact their reputation effect (U_R), but they also receive a transfer payment (T_R). In this circumstance, there is no opportunity for individuals to be excluded as members of society from receiving the subsidy. Indeed, the program increases the enforcer's total spending, although it does not affect the level of enforcement of the law. The new mixed strategy equilibrium of the game is given as follows:

$$\tilde{p}^* = \frac{C_E}{B_E + C_P - C_S} \qquad 21$$

$$\tilde{q}^* = \frac{U_O}{U_D + U_R + T_R} \qquad 22$$

> **Proposition 4**: The implementation of crime prevention initiatives reduces the probabilities of offending and of enforcing the law.

Obviously the result shows that $\tilde{p}^* < p*$ and $\tilde{q}^* < q^*$. This implies that the implementation of crime prevention initiatives decreases both the public's and the enforcer's probabilities of offending and of enforcing the law, respectively. It should be noted that the results hold given the levels of enforcement and punishment. The policy implication is that crime prevention initiatives are efficient in reducing individuals' offending behavior, provided that the authority does not reduce the levels of enforcement and punishment. Any attempts to reduce the level of either the enforcement or the severity of punishment, or both, in order to finance the implementation of crime prevention initiatives does not sustain the results above.

6. CONCLUSIONS

It has been shown in this study that both increasing the severity of punishment and initiating crime prevention programs may influence the offending behavior of individuals. The impact of the latter on reducing the probability of offending is more certain than the former, *ceteris paribus*. The results show that regardless of the interventions, the structure of the games does not change. The solutions of the games lie in a mixed strategy equilibrium, and thus a total reduction in the number of offenses is impossible.

From the perspective of the enforcer, implementing crime prevention initiatives is more appealing than any attempt to increase the severity of

punishment. The impact of the former on individual behavior is less ambiguous than that of the latter. One crucial factor which has not been captured in our analysis is the comparison of resources required to implement each policy. Crime prevention initiatives[19] may require considerably more resources than policies to increase the severity of punishment. In addition to that, it is implicitly assumed in our models that, if the law is enforced, then all incidents and offenses committed by individuals will be detected. In reality, due to various reasons,[20] this may not be the case, since there will always be offenses which remain undetected and unrecorded by the authority. Certainly this creates difficulties for the enforcer when differentiating between those who have committed an offense and those who have not. Indeed, while there is novelty in extending the analysis to incorporate this phenomenon, it may not be easy to do so from the game theoretical perspective.

Endnotes

* Centre for Criminal Justice Economics and Psychology, University of York, U.K. YO10 5DD. E-mail: rp104@york.ac.uk or rimawan@fe.ugm.ac.id. This article is part of my Ph.D. thesis. I am very grateful to two anonymous referees who provided useful comments. Furthermore, I am indebted to John Bone, Amanda Perry and Zoe Neale for helpful discussions, however, all remaining errors are my responsibility.

1. For instance, Ehrlich (1975, 1977), and Passell and Taylor (1977) among others. Recently, Levitt and Miles (2007) provide an excellent survey of empirical studies in this area.
2. See Estes (1944), Azrin (1959a, 1959b, and 1960), Appel (1961, 1963) and Storms et al. (1963) for food-motivated behavioral experiments by inducing electric shock in animals. See also Gneezy and Rustichini (2004), who conducted an experiment by imposing a small fine on parents late in picking up their children from day-care nursery.
3. This paper has inspired the development of crime economics. Surveys in the area have been conducted by several authors, including Garoupa (1997), Eide (2000, 2004), Bowles (2000) and Polinsky and Shavell (2000, 2007).
4. This article triggered a long debate which involved several authors, including BOT (1990), Weissing and Ostrom (1991), Hirshleifer and Rasmusen (1992) and Tsebelis (1990, 1991, 1992, 1993). Recently, Andreozzi (2004) attempted to accommodate both Tsebelis' approach and his critics by conducting the Stackelberg sequential analysis of the inspection game.
5. De Mesquita and Cohen (1995) analyzed a similar game in extensive form. Kilgour (1994) studied a similar game to investigate the effectiveness of enforcement when inspection is costless.

6. The paper consists of three parts written separately by William Bianco, Peter Ordeshook and George Tsebelis. Throughout this paper we refer to the article as BOT (1990).
7. Several studies which follow Becker's approach share this result, especially when they assume that government officers are corrupt (e.g., Bowles and Garoupa, 1997; Garoupa and Klerman, 2004; and Polinsky and Shavell, 2001). Bowles (2000) provides an excellent literature survey in this area.
8. For evidence in the U.K., see Bowles et al. (2004), Bowles and Pradiptyo (2004), Chapman et al. (2002) and DCA (2004). See Einat (2004) for evidence in Israel, and Tonry (1997) and U.S. DOJ (2003) for cases in the U.S.
9. This also applies for recent studies based on Becker's approach, for instance Garoupa (1997) and Polinsky and Shavell (2000, 2007), among others.
10. See Wittman (1985) for further discussion on the PIP.
11. For instance, when the game is played by more than 2 players or under imperfect information, when it is played sequentially or simultaneously, and when the game is played by rational or by unsophisticated players as in evolutionary game theory.
12. The social welfare function is defined as the sum of the offenders' expected utility from committing offenses, the harms caused by crime, and the costs of enforcing the law.
13. Under Becker's tradition, Garoupa and Klerman (2002) studied the deterrence effect under a rent-seeking government. Instead of maximizing the social welfare function, the government aims to maximize its own utility.
14. In many countries, for instance, individuals with a criminal history are not entitled to posts as police officers, soldiers, or civil servants. In addition, sexual offenders have been banned from posts which involve direct interaction with children.
15. This applies particularly to convicted pedophiles or sexual offenders.
16. Rasmusen (1996) found that stigma can deter individuals from having a criminal career. This applies particularly to non-convicted offenders. Funk (2004) studied the impact of stigma on both convicted and non-convicted offenders. For convicted offenders, stigma is effective if it is complemented by stronger formal punishment.
17. See Brand and Price (2000) and Dubourg et al. (2005) for an estimation of the social costs of crime in the U.K.
18. Einat (2004) reported that 12 other judges (30%) were of the view that fine enforcement is beyond their responsibility, and thus irrelevant.
19. For instance, the monitoring costs and/or the distribution costs of subsidies to those who do not have a criminal record.
20. For instance, victims and witnesses may be terrified or ashamed to report to police, no witnesses may have been present at the time the offenses were committed, or the offenses cannot be observed by police.

Review of Law & Economics, © 2007 by bepress

References

Andreozzi, L. 2004. "Rewarding Policemen Increases Crime. Another Surprising Result from the Inspection Game," 121 *Public Choice* 69-82.

Appel, J.B. 1961. "Punishment in the Squirrel Monkey *Saimiri Sciurea*," 133 *Science* 336.

_____. 1963. "Punishment and Shock Intensity," 141 *Science* 528-529.

Azrin, N.H. 1959a. "Some Notes on Punishment and Avoidance," 2 *Journal of the Experimental Analysis of Behavior* 260.

_____. 1959b. "Punishment and Recovery During Fixed-Ratio Performance," 2 *Journal of the Experimental Analysis of Behavior* 301-305.

_____. 1960. "Sequential Effects of Punishment," 131 *Science* 605-606.

Becker, G.S. 1968. "Crime and Punishment: An Economic Approach," 70 *Journal of Political Economy* 1-13.

Bianco, W.T., P.C. Ordeshook, and G. Tsebelis (BOT). 1990. "Crime and Punishment: Are One-Shot, Two-Person Games Enough?" 84 *American Political Science Review* 569-586.

Bowles, R.. 2000. "Corruption,". in B. Boudewijn and G. De Greest, eds. *Encyclopedia of Law and Economics*, *Vol.* 5, *The Economics of Crime and Litigation* 460-491. Edward Elgar.

_____ and N. Garoupa. 1997. "Casual Police Corruption and the Economics of Crime." 17 *Int'l Review of Law and Economics* 75-87.

_____, F. Gordon, R. Pradiptyo, C. McDougall, A. Perry, and R. Swaray. 2004. "Costs and Benefits of Sentencing Options – Report to the Home Office," unpublished manuscript, Centre for Criminal Justice Economics and Psychology, University of York.

_____ and R. Pradiptyo. 2004. "An Economic Approach to Offending, Sentencing and Criminal Justice Interventions – Report to Esmee Fairbairn Foundation," presented to Esmee Fairbairn Foundation, London.

Brand, S., and R. Price. 2000. "The Economic and Social Costs of Crime," *Home Office Research Series Paper* 217. London: Home Office.

Chapman, B., A. Mackie and J. Raine. 2002. "Fine Enforcement in Magistrates' Courts," *Home Office Development and Practice Report* 1, London, Home Office.

Clarke, S.H., and F.A. Campbell. 1998. "Can Early Intervention Prevent Crime Later? The Abecedarian Project Compared with Other Programs," 13 *Early Childhood Research Quarterly* 319-443.

Department for Constitutional Affairs (DCA). 2004. *Magistrates' Courts Business Returns–Annual Report* 2002-2003. London: Department for Constitutional Affairs.

Dubourg, R., J. Hamed, and J. Thorns. 2005. "The Economic and Social Costs of Crime Against Individuals and Households 2003/04," *Home Office Online Report* 30/05. London: Home Office.

Ehrlich, I. 1975. "The Deterrence Effect of Capital Punishment: A Question of Life and Death," 65 *American Economic Review* 397-417.

_____. 1977. "The Deterrence Effect of Capital Punishment: Reply," 67 *American Economic Review* 452-458.

Eide, E. 2000. "Economics of Criminal Behavior," in B. Boudewijn and G. De Greest, eds. *Encyclopedia of Law and Economics*, *Vol.* 5, *The Economics of Crime and Litigation* 345-389. Edward Elgar.

_____. 2004. "Recent Development in Economics of Crime," *German Working Papers in Law and Economics*, paper 8.

Einat, T. 2004. "Criminal Fine Enforcement in Israel; Administration, Policy, Evaluation and Recommendations," 6 *Punishment* & *Society* 175-194.

Estes, W.K. 1944. "An Experimental Study of Punishment," 57 *Psychological Monographs* No 3.

Funk, P. 2004. "On the Effective Use of Stigma as a Crime-Deterrent," 48 *European Economic Review* 715-728.

Garoupa, N. 1997. "The Theory of Optimal Law Enforcement." 11 *Journal of Economic Surveys* 267-295.

Garoupa, N., and D. Klerman. 2002. "Optimal Law Enforcement with a Rent-Seeking Government," 4 *American Law* & *Econ. Review* 116-140.

_____ and _____. 2004. "Corruption and the Optimal Use of Nonmonetary Sanctions," 24 *Int'l Review of Law* & *Economics* 219-225.

Gneezy, U., and A. Rustichini. 2004. "Incentives, Punishment and Behavior", in Camerer, Loewenstein and Rabin, eds. *Advances in Behavioral Economics*. Princeton Univ. Press.

Hirshleifer, J., and E. Rasmusen. 1992. "Are Equilibrium Strategies Unaffected by Incentives?" 4 *Journal of Theoretical Politics* 353-367.

Home Office. 2002. *The National Policing Plan* 2003-2006 (presented to Parliament by the Secretary of State for the Home Department in accordance with section 1 of the Police Reform Act 2002. London: Home Office.

Karoly, L.A., M.R. Kilburn., J.H. Bigelow., J.P. Caulkins., J.S. Cannon., and J.R. Chiesa. 2001. "Executive Summary: Assessing Costs and Benefits of Early Childhood Intervention Programs: Overview and Application to the Starting Early Starting Smart Program," Santa Monica: Rand Foundation. http://www.rand.org/pubs/monograph_reports/2005/MR1336.1.pdf.

Kilgour, D.M. 1994. "The Use of Costless Inspection in Enforcement," 36 *Theory and Decision* 207-232.

Levitt, S.D., and T.J. Miles. 2007 forthcoming. "Empirical Study of Criminal Punishment," in A.M. Polinsky and S. Shavell, eds. *Handbook of Law and*

Economics 1, North Holland.

De Mesquita, B., and L.E. Cohen. 1995. "Self Interest, Equity, and Crime Control: A Game-Theoretic Analysis of Criminal Decision Making," 33 *Criminology* 483-518.

Passell, P., and J.B. Taylor. 1977. "The Deterrence Effect of Capital Punishment: Another View," 67 *American Economic Review* 445-451.

Polinsky, A.M., and S. Shavell. 2000. "Economic Theory of Public Enforcement of Law," 38 *Journal of Economic Literature* 45-76.

_____ and _____. 2001. "Corruption and Optimal Law Enforcement." 81 *Journal of Public Economics* 1-24.

_____ and _____. 2007 forthcoming. "The Theory of Public Enforcement of Law," in A.M. Polinsky and S. Shavell, eds. *Handbook of Law and Economics* 1, North Holland.

Rasmusen, E. 1996. "Stigma and Self-Fulfilling Expectations of Criminality," 39 *Journal of Law and Economics* 519-544.

Storms, L.H., G. Boroczi, and W.E. Broen, Jr. 1963. "Punishment Inhibits an Instrument Response in Hooded Rats," 135 *Science* 1133.

Tonry, M. 1997. "Intermediate Sanctions in Sentencing Guidelines," *Issues and Practices in Criminal Justice*. National Institute of Justice, U.S. Dept. of Justice, http://www.ncjrs.gov/pdffiles/165043.pdf.

Tsebelis, G. 1989. "The Abuse of Probability in Political Analysis: The Robinson Crusoe Fallacy," 83 *American Political Science Revie*w 77-91.

_____. 1990. "Penalty Has No Impact on Crime? A Game Theoretical Analysis,".2 *Rationality and Societ*y 255-286.

_____. 1991. "The Effects of Fines on Regulated Industries: Game Theory vs. Decision Theory," 3 *Journal of Theoretical Politics* 81-101.

_____. 1992. "Are Sanctions Effective? A Game-Theoretic Analysis," 34 *Journal of Conflict Resolution* 3-28.

_____ 1993. "Penalty and Crime: Further Theoretical Considerations and Empirical Evidence," 5 *Journal of Theoretical Politics* 349-374.

U.S. Department of Justice (DOJ). 2003. "Felony Defendants in Large Urban Counties 2000." Washington DC: U.S. Department of Justice.

Weissing, F., and E. Ostrom. 1991. "Crime and Punishment: Further Reflections on the Counter-Intuitive Results of Mixed Equilibria Games," 3 *Journal of Theoretical Politics* 343-350.

Wittman, D. 1985. "Counter-Intuitive Results in Game Theory," 1 *European Journal of Political Economy* 77-89.

Youth Justice Board. 2003. "Evaluation of the Youth Inclusion Programme." London: http://www.youth-justice-board.gov.uk/Publications.

_____. 2005. "Monitoring and Evaluating the Safer School Partnership Programme." London: www.youth-justice-board.gov.uk/Publications.